

HARSHVARDHAN SINGH

AI / LLM Systems Engineer

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PROFESSIONAL SUMMARY

Applied AI engineer with 2+ years building production LLM systems — RAG pipelines, multi-provider routing, and on-prem Ollama inference — currently driving AI R&D at Accenture (JSW account) for Spring Boot platforms serving 15K+ users.

TECHNICAL SKILLS

AI / LLM: RAG (Retrieval-Augmented Generation), LangChain, LangChain4j, OpenAI, Anthropic Claude, Google Gemini, Ollama, Qwen, prompt engineering, function calling, agentic workflows, multi-model routing

Vector & Retrieval: FAISS, ChromaDB, embeddings, semantic chunking, hybrid retrieval; ML: Isolation Forest

Backend: Java, Python, Spring Boot, Spring Security, FastAPI, REST APIs, microservices, Hibernate, JPA

Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Apache Kafka, RabbitMQ

DevOps & Cloud: Docker, Kubernetes, GitHub Actions, Jenkins, AWS (EC2, S3, RDS), Nginx, Terraform

Frontend & Tools: Angular 17+, TypeScript, Git, JUnit, Mockito, Postman, Swagger / OpenAPI, Maven, Gradle

WORK EXPERIENCE

ACCENTURE

Pune, India

Associate Software Engineer — AI R&D Lead, JSW Account

Sep 2024 – Present

- Lead applied AI R&D on JSW account, productionizing RAG and LLM workflows across Spring Boot platforms serving 15K+ users.
- Architected RAG pipeline (Spring Boot, LangChain4j, FAISS, ChromaDB) over enterprise data; cut report turnaround ~80% and removed manual SQL reporting overhead via a schema-agnostic AI query layer.
- Migrated LLM workloads to on-prem Qwen / Ollama inference, eliminating data egress and cutting third-party API spend.
- Built AI support assistant with semantic retrieval over PDF / SOP corpora, deflecting routine queries 24x7 and shortening first-response time.
- Exposed retrieval workflows via JWT-secured REST APIs with prompt-level observability to debug hallucinations in production.

Software Engineer Intern

Feb 2024 – Jun 2024

- Refactored Spring Boot REST APIs and added JUnit / Mockito coverage on critical paths, reducing post-release issues ~15%.

AUTONMY — Independent AI Initiative

Founder & AI Engineer

2025 – Present

- Built AI Voice Receptionist MVP (LLM + voice pipeline) for SMB customer-call automation — appointment scheduling, query resolution, and conversational fallback in real time.
- Designed AI lead-generation and outreach automation for SMB sales — automated prospect discovery, personalized email generation, and multi-step follow-up sequencing.
- Productizing a Document RAG support assistant for SMB knowledge bases — semantic ingestion, conversational retrieval, and low-cost local inference.

PROJECTS

Smart LLM Routing Platform — Python, FastAPI, OpenAI, Anthropic, Ollama, Redis [Github](#)

- Designed multi-provider AI inference orchestration layer (OpenAI, Anthropic, Ollama) with cost, complexity, and token-aware routing for resilient, low-cost LLM operations.
- Built circuit-breaker provider health monitoring with exponential-backoff recovery, sub-second automatic failover, and structured per-request cost / latency telemetry for end-to-end observability.

AI-Driven Microservices Anomaly Detection — Spring Boot, Isolation Forest (ML), Redis [Github](#)

- Spring Boot backend ingesting microservice telemetry; Isolation Forest model flags latency anomalies in ~500 ms end-to-end.
- Cached feature windows in Redis to eliminate redundant model inference; exposed anomaly insights via REST for downstream dashboards and alerting.

EDUCATION

Madhav Institute of Technology and Science

Bachelor of Technology, Mechanical Engineering

Gwalior, India

2020 – 2024